

DEVOPS MASTER COURSE

Volume 7A

Microsoft Azure Foundations

Chapters 1-10

**AZURE | GLOBAL INFRASTRUCTURE | ENTRA ID | RBAC | ARM | VIRTUAL MACHINES |
VIRTUAL NETWORK | COST GOVERNANCE**

Beginner-to-enterprise Azure cloud engineering, hands-on labs, troubleshooting, governance,
and interviews

2026 Edition

About This Part

Volume 7A establishes the operating model required before an engineer deploys production workloads in Microsoft Azure. It begins with cloud and Azure fundamentals, then moves through global infrastructure, tenant and subscription hierarchy, administrative tools, identity, authorization, governance, compute, networking, and financial controls.

The central theme is controlled administration. A secure Azure environment separates human identities from workload identities, uses explicit scopes for access and policy, standardizes resource organization, designs for failure domains, and gives every resource an owner, purpose, lifecycle, and cost signal.

Scope and Safety

- Use a sandbox subscription, Azure free account, Azure for Students subscription, or an organization-approved training environment. Current eligibility, credits, regional service availability, quotas, and pricing must be verified before creating resources.
- Enable multifactor authentication, protect privileged roles, and use Microsoft Entra groups, Azure RBAC, managed identities, and just-in-time elevation instead of shared accounts or embedded secrets.
- Choose a permitted Azure region and verify availability-zone and service support. Region availability and service capabilities change over time.
- Apply budgets, expiry tags, and cleanup procedures before creating chargeable resources. Stopped virtual machines can continue to incur storage and networking charges unless they are deallocated and unused resources are removed.
- Do not expose SSH or RDP broadly to the internet. Prefer Azure Bastion, private connectivity, Just-in-Time access, or tightly restricted source addresses in approved environments.

How to Use the Labs

- Read the objective and design notes before opening the Azure portal or Cloud Shell.
- Use a consistent naming convention and tags such as owner, environment, workload, costCenter, dataClassification, and expiresOn.
- Capture command output, resource IDs, role assignments, policy compliance, screenshots, and cleanup evidence.
- Practice denial and failure scenarios deliberately: wrong subscription, wrong scope, missing role, disallowed region, closed port, and depleted quota.
- Rebuild successful manual labs later using Bicep or Terraform when the infrastructure-as-code volumes begin.

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VOLUME 7A | CHAPTER 1

Chapter 1 - Cloud Computing and Microsoft Azure

Understanding the operating model before choosing services

Learning Objectives

- Explain cloud computing through shared responsibility, elasticity, measured consumption, and automation.
- Compare IaaS, PaaS, SaaS, serverless, public cloud, private cloud, hybrid cloud, and multicloud.
- Describe Azure service families and the role of Azure in a DevOps delivery system.
- Identify workloads that should be rehosted, replatformed, refactored, retained, retired, or replaced.

1.1 Why Cloud Engineering Is an Operating Model

Cloud is not simply rented hardware. It changes how teams request capacity, govern access, deploy changes, observe health, recover from failure, and allocate cost. In a traditional data center, provisioning often depends on procurement and manual configuration. In Azure, an authenticated API request can create networks, compute, databases, identities, and policies in minutes. The speed creates business value only when security, standards, automation, and ownership are designed at the same time.

Why before how Learning a portal workflow without understanding lifecycle, identity, failure boundaries, and cost creates fragile environments. Professional cloud engineering starts with desired outcomes, constraints, and evidence.

1.2 Essential Cloud Characteristics

Characteristic	Engineering meaning	Operational implication
On-demand self-service	Authorized teams request resources without hardware procurement.	Guardrails and quotas must replace ticket-only control.
Broad network access	Services are reachable through documented network and API interfaces.	Identity, encryption, routing, and private endpoints become primary controls.
Resource pooling	Provider capacity is shared while tenants remain logically isolated.	Engineers design for quotas, noisy-neighbor risk, and service limits.
Rapid elasticity	Capacity can expand and contract with demand.	Applications should be stateless where possible and scaling must be tested.
Measured service	Usage is metered and billed.	Tagging, budgets, cost ownership, and optimization become engineering duties.
Automation and APIs	Resources are controlled through declarative or imperative interfaces.	Repeatability, review, and drift detection become achievable.

1.3 Service and Deployment Models

Model	Customer manages	Provider manages	Azure examples
IaaS	OS, middleware, applications, data, most configuration	Facilities, physical hosts, virtualization fabric	Azure Virtual Machines, Virtual Network
PaaS	Application code, data, service configuration	OS, runtime platform, patching of managed platform	Azure App Service, Azure SQL Database
Serverless	Functions or workflow logic, event contracts, data	Runtime provisioning and most scaling infrastructure	Azure Functions, Logic Apps
SaaS	Users, configuration, business data, access governance	Application and infrastructure stack	Microsoft 365, Dynamics 365

Public cloud uses provider-operated shared infrastructure. Private cloud dedicates a cloud-like environment to one organization. Hybrid cloud integrates on-premises and cloud environments. Multicloud deliberately uses services from multiple providers. Each adds integration, identity, networking, observability, and skills complexity; therefore it must be justified by business constraints rather than fashion.

1.4 Azure Service Families for DevOps

- Compute: Virtual Machines, Virtual Machine Scale Sets, App Service, Container Apps, Azure Kubernetes Service, and Functions.
- Networking: Virtual Network, Load Balancer, Application Gateway, Front Door, DNS, VPN Gateway, ExpressRoute, Firewall, Bastion, and Private Link.
- Storage and data: Storage accounts, managed disks, Azure Files, Azure SQL, Cosmos DB, PostgreSQL, MySQL, Cache for Redis, and analytics platforms.
- Identity and security: Microsoft Entra ID, Azure RBAC, managed identities, Key Vault, Defender for Cloud, Policy, and Microsoft Sentinel.
- DevOps and operations: Azure DevOps, GitHub, Azure Monitor, Log Analytics, Application Insights, Automation, Update Manager, and Resource Graph.
- Governance and cost: Management groups, subscriptions, tags, locks, Policy, Cost Management, Advisor, reservations, and savings plans.

1.5 Migration and Modernization Decisions

Strategy	Meaning	Suitable case
Rehost	Move with minimal architecture change.	Urgent data-center exit or compatible legacy workload.
Replatform	Adopt selected managed capabilities without redesigning the whole application.	Move database to managed service or replace self-hosted load balancer.
Refactor	Change architecture or code to use cloud-native patterns.	Scale, resilience, release speed, or cost requires deeper change.
Retain	Keep workload in current environment temporarily.	Dependency, regulation, latency, or migration risk blocks movement.
Retire	Decommission unused or duplicate workload.	No business owner, no use, or replacement already exists.
Replace	Adopt SaaS or a new product.	Maintaining custom system has less value than a supported service.

Hands-on Lab 1: Explore Azure and Build a Service Decision Record

Objective Create a structured first view of Azure and document why one workload would use IaaS, PaaS, or serverless.

Prerequisites

- Azure account with portal access or Microsoft Learn sandbox.
- A sample workload such as a small web application with database and file storage.
- A notes repository for architecture decision records.

Implementation

1. Open the Azure portal and identify the home page, global search, Cloud Shell, notifications, directory selector, subscription filter, and help menu.
2. Browse All services and classify at least fifteen services into compute, networking, storage, data, identity, security, monitoring, and governance.
3. For the sample workload, document business criticality, expected traffic, recovery objectives, data sensitivity, operating skills, and budget limit.
4. Create a one-page decision comparing a VM-based design, an App Service design, and a Functions-based design.
5. Record the chosen model, assumptions, rejected options, expected shared-responsibility tasks, and a review date.

Validation Evidence

- Portal screenshot with no sensitive identifiers.
- Service classification table.
- Architecture decision record with explicit assumptions and trade-offs.

Cleanup

- No chargeable resources are required. Remove any test dashboard tiles or temporary notes containing identifiers.

Stretch challenge Add a hybrid constraint: the database must remain on-premises. Redesign identity, network path, monitoring, and failure behavior.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Portal shows no subscriptions	Wrong directory, invitation not accepted, subscription disabled, or insufficient access	Check directory selector, subscriptions page, and account notifications	Switch tenant, accept invitation, contact subscription owner, or use a sandbox.
A service is unavailable	Region, subscription offer, quota, policy, or provider registration limitation	Check service availability, policy assignments, quotas, and resource providers	Select supported region/SKU, request quota, or register provider through approved process.

Symptom	Likely cause	Investigation	Corrective action
Design chooses a service only because it is managed	Requirements and exit criteria were not defined	Review availability, data, integration, skills, and cost assumptions	Create measurable decision criteria and compare operational responsibilities.

Chapter Summary

- Azure is an API-driven operating model, not merely hosted servers.
- Service models change the boundary of customer and provider responsibility.
- Migration strategy should be chosen per workload and business constraint.
- Every service decision must include identity, network, reliability, observability, security, and cost consequences.

Interview Questions and Model Answers

1. What is the main difference between IaaS and PaaS?

IaaS gives the customer more control and more operating responsibility, including guest OS and middleware. PaaS removes much of that platform administration so teams focus on application code, data, and service configuration.

2. Why can cloud speed increase risk?

APIs allow resources and privileges to be created quickly. Without policy, least privilege, logging, budgets, and automated standards, mistakes scale as rapidly as valid delivery.

3. When is rehosting appropriate?

When time, compatibility, or data-center exit urgency dominates and the workload can operate acceptably without immediate redesign. It should still have a later optimization plan.

4. What does shared responsibility mean?

Microsoft secures and operates the cloud platform according to the service model, while the customer remains responsible for areas such as identities, data, configuration, application security, and guest OS management where applicable.

Official Microsoft references: Azure training overview - <https://learn.microsoft.com/en-us/training/azure/> | Azure products and services - <https://azure.microsoft.com/en-us/products/> | Microsoft Cloud Adoption Framework - <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/>

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Chapter 2 - Azure Global Infrastructure and Reliability

Regions, availability zones, fault isolation, latency, and recovery design

Learning Objectives

- Distinguish geographies, regions, availability zones, region pairs, sovereign clouds, edge services, and fault domains.
- Select regions using compliance, latency, service availability, quota, resilience, and cost criteria.
- Compare zonal, zone-redundant, regional, and multi-region architecture patterns.
- Define RTO, RPO, service-level objectives, and failure tests.

2.1 Azure Geographies, Regions, and Zones

An Azure geography is a market and data-residency boundary that contains one or more regions. A region is a geographic area containing datacenters connected through a high-capacity network. Many regions provide availability zones: separate groups of datacenters with independent power, cooling, and network infrastructure. Zones reduce the probability that one localized failure removes every instance of a workload.

Critical distinction A region or zone is not a backup strategy. Availability protects service continuity; backup protects recoverability of data and configuration. Production systems normally need both.

Scope	Purpose	Design question
Geography	Data residency, compliance, and market boundary	Where may regulated data be stored and processed?
Region	Primary location for Azure resources	Which region provides required services, capacity, latency, and commercial terms?
Availability zone	Fault isolation inside a supported region	Which components must survive a datacenter-group failure?
Multi-region	Protection from regional disruption and global latency	What traffic, data, and operational model supports failover?
Edge/PoP services	Global entry, caching, routing, and protection	How should clients reach the closest healthy application endpoint?

2.2 Zonal and Zone-Redundant Resources

A zonal resource is pinned to a selected zone; the customer deploys enough instances across zones and manages distribution. A zone-redundant service replicates or distributes across zones as part of the managed service. The exact capability differs by service, SKU, and region, so support must be verified rather than assumed.

```

Client
|
Global or regional entry point
|
+----- Azure region -----+
| Zone 1: app instance   Zone 2: app instance |
| Zone 3: app instance   Zone-redundant data  |
+-----+

```

2.3 Region Selection Scorecard

Criterion	Evidence to collect	Common failure
Compliance and residency	Legal requirements, customer contracts, geography boundaries	Selecting cheapest region before confirming residency.
Service and feature availability	Service-by-region pages, supported SKUs, zone support	Assuming every Azure service is global.
Latency	Synthetic tests from users and dependencies	Choosing region based only on map distance.
Capacity and quota	Subscription quotas and SKU availability	Design passes review but cannot be deployed.
Resilience	Zone support, paired/secondary region plan, data replication	Using a single instance or single zone for critical service.
Cost	Compute, data transfer, operations, licensing, support	Comparing only VM price and ignoring network/managed-service cost.

2.4 RTO, RPO, and SLOs

Recovery time objective (RTO) is the maximum acceptable time to restore a business capability. Recovery point objective (RPO) is the maximum acceptable data loss measured in time. A service-level objective (SLO) is a measurable reliability target, such as successful request rate or latency. These values must be approved by the business because lower RTO and RPO normally increase architecture, testing, and operational cost.

Pattern	Typical objective	Trade-off
Single region, single instance	Development or disposable workloads	Lowest cost; no instance or zone resilience.
Single region, multi-zone	Regional high availability	Does not protect from a full regional disruption.
Active-passive multi-region	Defined disaster recovery	Requires replication, runbooks, capacity readiness, and regular tests.
Active-active multi-region	Global availability and latency	Highest data-consistency, routing, deployment, and incident complexity.

Hands-on Lab 2: Create a Region and Reliability Decision

Objective Produce a defensible region choice and failure-domain design for a customer-facing application.

Prerequisites

- A sample application and user geography.
- Access to Azure region and service availability documentation.
- Business-provided data residency, RTO, and RPO targets.

Implementation

6. Identify two candidate primary regions and one recovery region.
7. Check availability-zone support and required service/SKU availability for each candidate.
8. Document latency assumptions, user locations, on-premises dependencies, data-transfer paths, and quota risks.
9. Design a single-region multi-zone architecture and a multi-region recovery architecture.
10. Write failure tests for instance loss, zone loss, regional entry-point failure, data corruption, and operator error.
11. Create a decision record with review triggers such as new regulation, service retirement, or customer expansion.

Validation Evidence

- Completed region scorecard.
- Architecture flow showing failure boundaries.
- RTO/RPO mapping and test schedule.

Cleanup

- No resources are required for this design lab.

Stretch challenge Estimate the operational difference between active-passive and active-active designs, including deployment coordination and data conflict handling.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Required SKU not available	Regional capacity or subscription quota	Check SKU availability and quota pages	Choose alternative size/region or request quota before migration date.
Zone identifiers differ across subscriptions	Physical zone mapping is subscription-specific	Review logical/physical zone guidance for cross-subscription designs	Use supported mapping methods and avoid assuming zone 1 is physically identical.
Failover test meets infrastructure RTO but application remains unavailable	DNS, identity, secrets, dependencies, or data were not included	Trace full request and dependency chain	Test business transaction recovery, not only resource health.

Chapter Summary

- Regions provide deployment locations; availability zones provide fault isolation within supported regions.
- Zonal and zone-redundant behavior differs by service and SKU.
- Region selection requires compliance, service, latency, capacity, resilience, and cost evidence.
- RTO, RPO, SLOs, and failure tests convert reliability goals into engineering requirements.

Interview Questions and Model Answers

1. What is an Azure availability zone?

A separated group of datacenters within a region with independent power, cooling, and network infrastructure, used to isolate localized failures.

2. Does deploying to three zones automatically make an application highly available?

No. The application, load distribution, state, data, dependencies, identity, and operations must all survive and recover from failure.

3. What is the difference between RTO and RPO?

RTO measures acceptable restoration time; RPO measures acceptable data loss expressed as time since the last recoverable point.

4. Why verify service availability by region?

Azure capabilities, SKUs, availability-zone support, quotas, and previews vary by region and can change over time.

Official Microsoft references: Azure regions overview - <https://learn.microsoft.com/en-us/azure/reliability/regions-overview> | List of Azure regions - <https://learn.microsoft.com/en-us/azure/reliability/regions-list> | Azure availability zones overview - <https://learn.microsoft.com/en-us/azure/reliability/availability-zones-overview> | Reliability documentation - <https://learn.microsoft.com/en-us/azure/reliability/>

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Chapter 3 - Tenants, Management Groups, Subscriptions, and Resource Groups

Designing the administrative and billing hierarchy

Learning Objectives

- Explain Microsoft Entra tenants, billing scopes, management groups, subscriptions, resource groups, resources, and resource providers.
- Design a subscription and management-group hierarchy for environments and business ownership.
- Use resource groups based on lifecycle rather than organizational convenience alone.
- Identify scope inheritance for Azure RBAC and Azure Policy.

3.1 The Azure Resource Hierarchy

```

Microsoft Entra tenant
|
Tenant root management group
+-- Platform management group
|   +-- Connectivity subscription
|   +-- Identity subscription
+-- Landing Zones management group
|   +-- Production subscription
|       +-- Resource groups
|       +-- Resources
+-- Nonproduction subscription
  
```

Most Azure resources are deployed into resource groups, resource groups are contained in subscriptions, subscriptions can be organized under management groups, and the hierarchy is associated with a Microsoft Entra tenant. Access and policy assigned at a higher scope can be inherited by lower scopes, which makes hierarchy design a security and governance decision.

Scope	Primary purpose	Key design rule
Microsoft Entra tenant	Identity and directory boundary	Do not confuse tenant roles with Azure resource roles.
Management group	Govern subscriptions at scale	Keep hierarchy understandable; design for policy and delegation.
Subscription	Billing, quota, deployment, and access boundary	Separate workloads/environments when ownership, risk, or limits differ.
Resource group	Lifecycle and management container	Group resources that are deployed, updated, and removed together.
Resource	Managed service instance	Use consistent naming, region, tags, diagnostics, and ownership.
Resource provider	Namespace exposing a resource type	Registration and API versions affect deployment availability.

3.2 Subscription Design

Subscriptions are not merely invoices. They provide strong boundaries for quotas, policy, access delegation, and operational blast radius. Enterprises commonly use separate subscriptions for platform services, connectivity, identity, management, production workloads, nonproduction workloads, sandbox use, and regulated workloads. The exact hierarchy must remain simple enough to operate.

Separation driver	Example	Benefit
Environment	Production vs nonproduction	Reduces accidental cross-environment access and quota contention.
Business ownership	Retail vs manufacturing	Clear cost and service ownership.
Regulation/data class	Restricted workload subscription	Stronger policy and evidence boundary.
Platform function	Connectivity or management subscription	Central services can be delegated to platform teams.
Scale/quota	High-scale workload in separate subscription	Avoids one workload consuming shared limits.

3.3 Resource Group Lifecycle Principle

A resource group should usually contain resources that share a lifecycle and administrative context. Deleting a resource group initiates deletion of contained resources, subject to locks and service-specific dependencies. Placing a long-lived shared network and a short-lived application in the same resource group can create risky cleanup and ownership conflicts.

3.4 Naming and Tagging Foundations

Element	Example	Purpose
Resource group	rg-payments-prod-cin-001	Type, workload, environment, region code, sequence.
Virtual network	vnet-payments-prod-cin-001	Human-readable and automation-friendly identity.
Tags	owner=team-payments; environment=prod; expiresOn=never	Search, cost allocation, automation, and accountability.
Resource ID	/subscriptions/.../resourceGroups/.../ providers/...	Canonical ARM scope used by APIs and RBAC.

Hands-on Lab 3: Create and Inspect the Azure Resource Hierarchy

Objective Create a governed resource group, inspect resource IDs and providers, and practice subscription context.

Prerequisites

- Azure subscription with permission to create and delete resource groups.
- Azure Cloud Shell or locally installed Azure CLI.
- An approved region such as centralindia or another permitted location.

Implementation

12. Run `az account list --output table` and identify the active subscription and tenant.
13. Set the intended subscription with `az account set --subscription "<subscription-id-or-name>"`.
14. Create a resource group: `az group create --name rg-v7a-foundation-lab --location centralindia --tags owner=<name> environment=lab expiresOn=<date>`.
15. Inspect the resource group with `az group show --name rg-v7a-foundation-lab --output jsonc`.
16. List registered providers with `az provider list --query "[?registrationState=='Registered'].namespace" --output table`.
17. Locate the resource-group resource ID and explain each path segment.
18. Add a second tag and verify it in the portal and CLI.

Validation Evidence

- Output showing intended subscription and tenant.
- Resource-group JSON including location, tags, and ID.
- Provider registration list and a written explanation of scope inheritance.

Cleanup

- Delete the group with `az group delete --name rg-v7a-foundation-lab --yes --no-wait` after later chapters no longer depend on it.

Stretch challenge Propose a management-group and subscription hierarchy for a company with central platform, three business units, production/nonproduction, and one regulated workload.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Resource created in wrong subscription	CLI context was not checked	Run <code>az account show</code> and inspect resource ID	Set subscription explicitly and include context validation in scripts.
Resource type cannot be created	Provider not registered, policy denial, quota, or regional support	Inspect deployment error, provider registration, policy compliance, and quotas	Register provider through approved process or correct policy/region/quota issue.
Deleting resource group fails	Lock, backup item, dependency, or protected service	List locks and inspect deployment operations	Remove approved blockers in correct order and preserve required backup evidence.

Chapter Summary

- Azure resource scopes form an inheritance hierarchy associated with a Microsoft Entra tenant.
- Subscriptions are access, quota, governance, billing, and blast-radius boundaries.
- Resource groups should reflect shared lifecycle and administrative context.
- Resource IDs and provider namespaces are essential for CLI, policy, RBAC, and automation.

Interview Questions and Model Answers

1. What is the difference between a tenant and a subscription?

A tenant is the Microsoft Entra identity and directory boundary. A subscription is an Azure resource, quota, billing, and governance boundary associated with a tenant.

2. Why use management groups?

They organize subscriptions and allow consistent Azure Policy and RBAC assignment at scale.

3. How should resources be grouped into resource groups?

Prefer resources that share lifecycle, ownership, environment, and management context. Avoid mixing disposable application resources with long-lived shared infrastructure.

4. What happens when a role is assigned at subscription scope?

The assignment can be inherited by resource groups and resources under that subscription unless a more specific control changes effective access.

Official Microsoft references: Azure management groups - <https://learn.microsoft.com/en-us/azure/governance/management-groups/overview> | Azure Resource Manager overview - <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/overview> | Organize Azure resources - <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/azure-setup-guide/organize-resources> | Naming rules and restrictions - <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/resource-name-rules>

VOLUME 7A | CHAPTER 4

Chapter 4 - Azure Portal, Cloud Shell, CLI, PowerShell, APIs, and SDKs

Moving from exploration to repeatable administration

Learning Objectives

- Choose the Azure portal, Cloud Shell, Azure CLI, Azure PowerShell, REST APIs, SDKs, or Infrastructure as Code for a task.
- Authenticate safely and set subscription context explicitly.
- Use Azure CLI output formats, JMESPath queries, help, and idempotent scripting practices.
- Explain Azure Resource Manager as the common management plane.

4.1 One Management Plane, Multiple Interfaces

The Azure portal, CLI, PowerShell, SDKs, REST clients, Bicep, and Terraform ultimately interact with Azure management APIs and resource providers. The portal is excellent for discovery and visual inspection, but repeatable production changes should be reviewable, parameterized, logged, and preferably declared as code.

Interface	Best use	Risk to manage
Azure portal	Discovery, one-time inspection, visual troubleshooting	Manual drift and undocumented changes.
Cloud Shell	Authenticated browser terminal with Bash or PowerShell	Session limits, temporary host, and dependency on browser access.
Azure CLI	Cross-platform interactive work and scripting	Wrong subscription, parsing fragile human-readable output, user-auth automation.
Azure PowerShell	PowerShell-oriented administration and object pipelines	Module version drift and implicit context.
REST/SDK	Applications, tooling, and platform integrations	Token handling, retries, API versions, and rate limits.
Bicep/Terraform	Repeatable reviewed infrastructure deployment	State, module quality, secrets, and change governance.

4.2 Azure Cloud Shell

Cloud Shell is an authenticated browser-accessible terminal with Bash and PowerShell choices and common Azure tools installed. It is useful for labs, incident access, and administration from managed browsers. It is not a general-purpose production compute platform. Files in the home directory can be persisted through the configured storage experience, while the underlying session host is temporary and times out after inactivity.

4.3 Azure CLI Fundamentals

```
az login
az account list --output table
az account show --output jsonc
az account set --subscription "<subscription>"
az group list --query "[].{name:name,location:location}" --output table
az find vm
az vm create --help
```

Human users should authenticate interactively with modern Microsoft Entra controls. Automated workloads should use workload identities such as managed identities, federated credentials, or carefully governed service principals rather than username/password scripts or copied access tokens.

4.4 Script Quality Controls

- Fail fast when tenant, subscription, region, or required variables are wrong.
- Use structured output such as JSON and explicit queries rather than parsing decorative tables.
- Make scripts idempotent or detect existing state before creating resources.
- Log commands, correlation IDs, deployment names, and resource IDs without logging secrets.
- Use what-if or preview capabilities for declarative deployments and require review for destructive changes.
- Pin or test tool versions in CI and review breaking-change notices.

4.5 ARM Operations and Activity Log

Azure Resource Manager provides the management layer used to create, update, and delete resources. Management operations produce deployment and activity evidence that helps troubleshoot authorization failures, policy denials, invalid properties, dependencies, quotas, and provider errors. Data-plane operations, such as reading a blob or querying a database, can use different endpoints and authorization models.

Hands-on Lab 4: Operate Azure with Cloud Shell and CLI

Objective Use Azure CLI safely, query structured data, and create a repeatable resource-group workflow.

Prerequisites

- Azure subscription and Cloud Shell access.
- Permission to create and delete resource groups.
- A naming prefix unique to your account.

Implementation

19. Open Cloud Shell and choose Bash or PowerShell.
20. Run `az version` and record the Azure CLI and extension versions.
21. Run `az account show` and confirm tenant, subscription, and user identity.
22. Create variables for location and resource-group name.
23. Create the resource group with tags, then query only name, location, provisioningState, and tags using `--query`.
24. Export the resource-group JSON to a file and open it in the Cloud Shell editor.
25. List recent activity with `az monitor activity-log list --resource-group <name> --max-events 20 --output table`.
26. Delete the resource group and verify it no longer appears.

Validation Evidence

- CLI version output.
- Context validation output.
- JSON evidence and activity-log evidence.
- Cleanup confirmation.

Cleanup

- Remove the Cloud Shell files if they contain identifiers not required for your portfolio.

Stretch challenge Write a shell function that refuses to continue unless the active subscription ID equals an expected value and the environment tag is lab.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
az command not found	CLI not installed or shell PATH not refreshed	Run <code>az --version</code> and inspect installation	Use Cloud Shell or install supported Azure CLI and reopen terminal.
Authentication succeeds but commands fail	Wrong tenant/subscription or missing role	Run <code>az account show</code> and inspect effective access	Switch context or request least-privilege assignment at correct scope.
Script broke after CLI update	Command output or parameter behavior changed	Compare versions and breaking-change notes	Pin/test versions and use structured output with automated tests.
AuthorizationFailed despite recent role assignment	Token cache or propagation delay	Refresh login and verify assignment scope	Reauthenticate, wait for propagation, and inspect deny assignments/policy.

Chapter Summary

- Azure interfaces converge on Resource Manager and service APIs.
- Portal use is valuable for exploration; production changes should be repeatable and reviewable.
- CLI automation must validate context and use workload identities.
- Structured output, idempotency, logging, and version testing make scripts reliable.

Interview Questions and Model Answers

1. When would you choose Azure CLI over the portal?

For repeatable administration, scripting, CI/CD integration, structured queries, and operations that must be logged or reproduced.

2. What is Azure Cloud Shell?

An authenticated browser-accessible Bash or PowerShell environment with Azure tools preinstalled, intended for Azure management rather than general compute.

3. Why is `az account show` important?

It confirms the active tenant and subscription so commands do not modify the wrong environment.

4. What is the difference between management plane and data plane?

The management plane creates and configures resources through ARM and resource providers. The data plane performs service-specific work such as reading objects, sending messages, or querying data.

Official Microsoft references: Azure Cloud Shell overview - <https://learn.microsoft.com/en-us/azure/cloud-shell/overview> | Azure CLI documentation - <https://learn.microsoft.com/en-us/cli/azure/> | Get started with Azure CLI - <https://learn.microsoft.com/en-us/cli/azure/get-started-with-azure-cli> | Azure Resource Manager overview - <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/overview>

VOLUME 7A | CHAPTER 5

Chapter 5 - Microsoft Entra ID Fundamentals

Identity as the control plane for people, applications, and devices

Learning Objectives

- Explain tenants, users, groups, applications, service principals, devices, external identities, and authentication methods.
- Distinguish Microsoft Entra roles from Azure resource roles.
- Design administrator separation, MFA, Conditional Access, break-glass access, and lifecycle controls.
- Identify human and workload authentication patterns.

5.1 Identity Is the First Security Boundary

Microsoft Entra ID is Microsoft's cloud identity and access service. Azure subscriptions trust an Entra tenant for identities. An attacker with privileged identity access can create credentials, assign roles, disable controls, or access resources even when network boundaries are well designed. Therefore identity hardening precedes workload deployment.

Object	Purpose	Example
User	Represents a human identity	Engineer, administrator, external collaborator.
Group	Collects identities for scalable access	Platform-Operators, Billing-Readers.
Application registration	Defines an application identity and configuration	Internal deployment application.
Service principal	Tenant-local instance of an application identity	CI/CD workload granted Azure role.
Managed identity	Azure-managed workload identity	VM or Function authenticating to Key Vault.
Device	Registered or joined endpoint identity	Managed Windows workstation.
External identity	Guest or partner collaboration identity	Consultant from another organization.

5.2 Authentication Controls

- Multifactor authentication reduces the value of a stolen password.
- Passwordless methods reduce phishing and password-reuse risk when implemented and recovered securely.
- Conditional Access evaluates signals such as user, role, application, device, location, and risk before granting access.
- Single sign-on centralizes authentication and can simplify revocation and monitoring.
- Emergency access accounts provide a controlled recovery path when normal federation or policy fails; they require strong monitoring and documented tests.

5.3 Entra Roles vs Azure Roles

Role system	Controls	Example roles
Microsoft Entra roles	Directory objects and identity services	Global Administrator, User Administrator, Application Administrator.
Azure RBAC roles	Azure resources at ARM scopes	Reader, Contributor, Owner, Virtual Machine Contributor.
Service/data roles	Service-specific data-plane operations	Storage Blob Data Reader, Key Vault Secrets User.

Common mistake Giving someone Global Administrator does not automatically make that person Owner of every subscription. Directory administration and Azure resource authorization are separate systems, although privileged users may be able to elevate or influence access depending on controls.

5.4 Identity Lifecycle

Joiner, mover, and leaver processes must create, modify, and remove access according to employment state and business need. Access should flow through groups and eligible privileged roles wherever possible. Reviews should detect stale guests, dormant service principals, expired owners, unused credentials, and direct assignments that bypass the access model.

5.5 Workload Identity Strategy

Applications need identity but should not imitate users. Preferred patterns are managed identities for Azure-hosted workloads and workload identity federation for external CI/CD platforms. Client secrets and certificates can still be required in some systems, but they need controlled storage, expiry, rotation, and ownership.

Hands-on Lab 5: Build an Identity and Administrator Control Plan

Objective Design Microsoft Entra identity controls for a small engineering organization without creating unnecessary privileged access.

Prerequisites

- Access to view Microsoft Entra ID or Microsoft Learn identity modules.
- A fictional organization with platform, application, security, finance, and external support teams.
- No Global Administrator privileges are required for the design exercise.

Implementation

27. List the human identity populations, workload identities, devices, guests, and emergency access identities.
28. Define groups for job functions and environments; avoid assigning permissions to individuals in the target state.
29. Map required Microsoft Entra roles separately from Azure resource roles.
30. Define MFA, Conditional Access, privileged workstation, approval, and access-review expectations.
31. Design joiner, mover, leaver, guest-expiry, credential-expiry, and emergency-access testing processes.
32. Create a diagram showing authentication, authorization, and audit paths.

Validation Evidence

- Identity inventory and group model.
- Privilege matrix separating Entra and Azure roles.
- Lifecycle and emergency-access runbook.

Cleanup

- Do not create privileged accounts solely for a lab. Remove any temporary guest invitations or test applications if created in an approved sandbox.

Stretch challenge Add a CI/CD platform outside Azure. Choose between service principal secret, certificate, or federated credential and justify the risk decision.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
User can sign in but cannot manage resource	Authentication succeeded but Azure RBAC is missing	Check subscription/resource IAM and effective role assignments	Assign minimum Azure role at correct scope through group or eligible role.
Administrator blocked by Conditional Access	Policy condition, device state, location, or authentication method	Use sign-in logs and policy result details	Correct device/authentication state or follow emergency access process; do not broadly disable policy.
Application authentication fails	Wrong tenant, audience, credential, consent, or service principal	Inspect token claims, app registration, credential expiry, and sign-in logs	Correct configuration and prefer managed/federated identity where possible.
Departed user still has access	Direct assignment, guest account, cached session, or unmanaged app credential	Run access review and inspect assignments/credentials	Revoke sessions, remove assignments, disable/delete identity, rotate affected credentials.

Chapter Summary

- Microsoft Entra ID is the identity boundary trusted by Azure subscriptions.
- Users, groups, service principals, managed identities, devices, and guests have different lifecycle and security needs.
- Microsoft Entra roles and Azure RBAC roles control different scopes.
- MFA, Conditional Access, privilege separation, access reviews, and emergency access are foundational controls.

Interview Questions and Model Answers

1. What is a Microsoft Entra tenant?

A dedicated identity and directory instance containing users, groups, applications, devices, policies, and related identity configuration.

2. What is a service principal?

A tenant-local security identity representing an application or service and used for authentication and authorization.

3. What is the difference between an application registration and a service principal?

The application object defines the application globally in its home tenant; a service principal is the local instance used in a tenant for sign-in and access.

4. Why prefer groups for access?

Groups make assignment, review, onboarding, role change, and removal more scalable and auditable than individual direct assignments.

Official Microsoft references: Microsoft Entra documentation - <https://learn.microsoft.com/en-us/entra/> | Azure identity, access, and security fundamentals - <https://learn.microsoft.com/en-us/training/modules/describe-azure-identity-access-security/> | Managed identities overview - <https://learn.microsoft.com/en-us/entra/identity/managed-identities-azure-resources/overview> | Conditional Access documentation - <https://learn.microsoft.com/en-us/entra/identity/conditional-access/>

VOLUME 7A | CHAPTER 6

Chapter 6 - Azure RBAC, Managed Identities, and Privileged Access

Authorizing the minimum action at the minimum scope for the minimum time

Learning Objectives

- Explain role definitions, role assignments, security principals, scopes, inheritance, deny assignments, and effective access.
- Compare system-assigned and user-assigned managed identities.
- Use PIM concepts for eligible, time-bound, approved privileged access.
- Design custom roles and data-plane roles cautiously.

6.1 Azure RBAC Model

An Azure role assignment combines three elements: a security principal, a role definition, and a scope. The resulting permission is inherited by lower scopes. Effective access can also be influenced by deny assignments, policy, service-specific authorization, network restrictions, and tenant controls.

Principal (who) + Role definition (what) + Scope (where) = Role assignment

Built-in role	Capabilities	Caution
Reader	View Azure resources and configuration	Does not necessarily grant data-plane read access.
Contributor	Manage resources but not grant Azure RBAC access	Still broad; prefer service-specific contributor roles.
Owner	Manage resources and grant access	Highly privileged; use eligible, time-bound assignment.
User Access Administrator	Manage user access to Azure resources	Can create powerful role assignments.
Service-specific roles	Manage selected resource type or data operations	Select management-plane vs data-plane role intentionally.

6.2 Scope and Inheritance

Assignments can be made at management group, subscription, resource group, or resource scope. Higher-scope assignment is easier to administer but increases blast radius. Resource-level assignment is precise but can create unmanageable complexity. A common enterprise pattern assigns broad read access at higher scope, operational roles at subscription or resource-group scope, and exceptional data access at resource scope.

6.3 Managed Identities

Type	Lifecycle	Use case
System-assigned	Created and deleted with one Azure resource	One workload needs one identity with matching lifecycle.

Type	Lifecycle	Use case
User-assigned	Independent Azure resource attachable to multiple workloads	Shared identity, preauthorization, or identity lifecycle independent of compute.

Managed identities remove the need to store credentials in code. The workload requests a token from the Azure identity platform and uses that token with a service that supports Microsoft Entra authentication. Managed identity does not automatically grant permission; an Azure or service role assignment is still required.

6.4 Privileged Identity Management

Microsoft Entra Privileged Identity Management supports just-in-time and time-bound access for important roles. An eligible user activates a role when needed and can be required to provide MFA, justification, approval, and a limited duration. PIM activity and access reviews reduce permanent privilege and stale assignments.

6.5 Custom Role Discipline

- Start with built-in roles and document the exact missing actions before creating a custom role.
- Separate role definition from assignment scope.
- Avoid wildcard permissions that silently expand when services add actions.
- Version custom roles as code and review changes like application code.
- Test positive and negative actions and inspect effective access.
- Create an owner, review date, and deprecation process for every custom role.

Hands-on Lab 6: Assign Least-Privilege Access to a Managed Identity

Objective Create a user-assigned managed identity and grant it limited read access to one resource group.

Prerequisites

- Permission to create a resource group and user-assigned managed identity.
- Permission to create role assignments in an approved sandbox, or a supervisor-provided assignment.
- Azure CLI in Cloud Shell.

Implementation

33. Create a resource group named `rg-v7a-identity-lab` in an approved region.
34. Create a user-assigned identity: `az identity create --resource-group rg-v7a-identity-lab --name id-v7a-reader`.
35. Capture the `principalId` and resource ID from `az identity show`.
36. Assign Reader at the resource-group scope using `az role assignment create --assignee-object-id <principalId> --assignee-principal-type ServicePrincipal --role Reader --scope <resource-group-id>`.
37. List role assignments for the identity and explain the principal, role, and scope.
38. Document why Contributor, Owner, and subscription scope would be excessive.
39. If available, review PIM settings for a privileged Azure role without activating it.

Validation Evidence

- Identity JSON with principal ID.
- Role assignment output and scope.
- Least-privilege justification and cleanup evidence.

Cleanup

- Delete the role assignment, managed identity, and resource group after validation.

Stretch challenge Design a separation-of-duties model where platform engineers deploy resources but only security administrators can change policy and privileged role settings.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Managed identity receives 403	Role missing, wrong data-plane role, wrong scope, token audience, or propagation	Inspect role assignments and service authorization model	Assign correct minimum role, refresh token, and wait for propagation.
Role assignment command is denied	Caller lacks role-assignment permission or deny assignment applies	Check effective access and activity log	Use authorized access workflow; do not grant Owner merely to bypass process.
User is eligible but cannot activate role	PIM settings require MFA, approval, justification, or valid schedule	Review activation error and role settings	Meet requirements or contact configured approver.
Custom role works too broadly	Wildcard actions or assignableScopes are excessive	Review role JSON and negative tests	Restrict actions/notActions/dataActions and narrow assignable scopes.

Chapter Summary

- Azure RBAC combines principal, role, and scope.
- Managed identities solve credential management, not authorization by themselves.
- PIM reduces standing privilege with eligible and time-bound activation.
- Custom roles require code review, negative tests, ownership, and lifecycle management.

Interview Questions and Model Answers

1. What are the three components of an Azure role assignment?

A security principal, a role definition, and a scope.

2. What is the difference between system-assigned and user-assigned managed identity?

System-assigned identity shares the lifecycle of one resource. User-assigned identity is independent and can be attached to multiple resources.

3. Why might Reader not allow reading application data?

Reader is generally a management-plane role. Services can require separate data-plane roles such as Storage Blob Data Reader.

4. What problem does PIM solve?

It reduces standing privileged access by making roles eligible, time-limited, auditable, and optionally subject to MFA, justification, and approval.

Official Microsoft references: Azure RBAC overview - <https://learn.microsoft.com/en-us/azure/role-based-access-control/overview> | Azure built-in roles - <https://learn.microsoft.com/en-us/azure/role-based-access-control/built-in-roles> | Managed identities overview - <https://learn.microsoft.com/en-us/entra/identity/managed-identities-azure-resources/overview> | Privileged Identity Management overview - <https://learn.microsoft.com/en-us/entra/id-governance/privileged-identity-management/pim-configure>

VOLUME 7A | CHAPTER 7

Chapter 7 - Azure Resource Manager Governance: Tags, Locks, and Policy

Standardizing resource state while preserving delivery speed

Learning Objectives

- Explain Resource Manager, resource providers, deployments, templates, dependencies, and scopes.
- Use tags and naming conventions for ownership, automation, security, and cost.
- Apply Delete and Read-only locks appropriately.
- Explain Azure Policy definitions, initiatives, assignments, effects, exemptions, compliance, and remediation.

7.1 Resource Manager as the Management Layer

Azure Resource Manager receives management requests, authenticates and authorizes them, evaluates applicable governance, and routes operations to resource providers. Declarative deployments describe a target resource state and dependencies. Bicep is Azure's domain-specific language for authoring ARM deployments and will be covered deeply in a later volume.

7.2 Tags Are Metadata, Not Security Boundaries

Tag	Example	Operational use
owner	team-payments	Escalation and accountability.
environment	prod	Policy, automation, dashboards, and cost filtering.
costCenter	FIN-204	Chargeback/showback and budget ownership.
dataClassification	confidential	Security workflow input; not a substitute for access control.
expiresOn	2026-09-30	Automated cleanup candidate.
managedBy	bicep-platform-v2	Drift and deployment ownership.

Tag support and inheritance behavior vary across resource types. Azure Policy can add or require tags, and Cost Management offers additional allocation capabilities, but engineers must test actual propagation and billing behavior.

7.3 Resource Locks

A Delete/CanNotDelete lock prevents deletion while permitting modification. A Read-only lock prevents updates and deletion and can disrupt service operations that internally require write actions. Locks override user permissions at the protected scope and are inherited. They are valuable for critical shared resources, but they require documented ownership and emergency-removal procedures.

7.4 Azure Policy

Azure Policy evaluates resource state against organizational rules. A policy definition contains a condition and an effect. Definitions can be grouped into initiatives and assigned at management-group, subscription, resource-group, or resource scope. Common effects include audit, deny, modify, deployIfNotExists, append, disabled, and other service-specific behaviors. Remediation uses managed identity permissions when policy must modify or deploy resources.

Control	Primary question	Example
Azure RBAC	Who may perform an action at a scope?	Platform group can create VNets in a subscription.
Azure Policy	What resource state is permitted or required?	Only approved regions and SKUs; diagnostic settings required.
Resource lock	Can this scope be deleted or modified?	Shared DNS zone cannot be deleted accidentally.
Tagging standard	Who owns it and how is it classified?	Every resource has owner and environment tags.

7.5 Policy Rollout Strategy

40. Start with inventory and audit effects to understand existing noncompliance.
41. Confirm ownership, exclusions, unsupported resource behavior, and remediation impact.
42. Use initiatives and management-group scope for consistent enterprise standards.
43. Pilot deny/modify behavior in sandbox and nonproduction subscriptions.
44. Create time-bound exemptions with business owner, compensating control, and expiry.
45. Monitor compliance drift and remediate through approved automation.

Hands-on Lab 7: Apply Tags, a Delete Lock, and an Audit Policy

Objective Build a small governance baseline and observe how different controls behave.

Prerequisites

- Permission to create a resource group, lock, and policy assignment in an approved sandbox.
- Azure CLI or portal.
- A permitted test region.

Implementation

46. Create resource group rg-v7a-governance-lab with owner, environment, costCenter, and expiresOn tags.
47. Create a CanNotDelete lock: az lock create --name protect-lab --lock-type CanNotDelete --resource-group rg-v7a-governance-lab.
48. Attempt to delete the resource group and capture the failure.
49. Locate a built-in policy that audits a missing tag or restricts allowed locations; assign it at the resource-group scope using Audit where possible.
50. Create or inspect a noncompliant test condition and review compliance results after evaluation.
51. Remove the lock through the approved process and document why locks and RBAC solve different problems.
52. Remove the policy assignment and delete the resource group.

Validation Evidence

- Tag output, lock output, and denied deletion evidence.
- Policy assignment and compliance screenshot/output.
- Cleanup confirmation.

Cleanup

- Remove policy assignment before deleting the lab resource group. Confirm no remediation identity or assignment remains.

Stretch challenge Design a policy initiative for required diagnostics, approved regions, allowed VM SKUs, public IP restrictions, and required tags. Define staged rollout and exemption rules.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Owner cannot delete resource	Inherited lock overrides permissions	List locks at resource, resource group, and subscription	Remove authorized lock after impact review.
Policy assignment shows noncompliance but does not fix resource	Audit effect or missing remediation identity/permissions	Inspect effect, remediation task, and identity role	Use modify/deployIfNotExists correctly and grant minimum remediation permissions.
Valid deployment denied	Policy parameters, alias behavior, or inherited initiative	Review policy events and compliance reason	Correct resource or create approved time-bound exemption; avoid disabling enterprise policy globally.
Tags missing from cost report	Resource does not emit tags as expected or reporting delay	Check resource support, inheritance, and cost dimensions	Use supported tag inheritance/cost allocation and standardize at provisioning time.

Chapter Summary

- Resource Manager is Azure's management layer and scope model.
- Tags provide metadata; locks protect against modification/deletion; RBAC controls actor permissions; Policy controls resource state.
- Policy should be rolled out from audit to enforcement with pilots and time-bound exemptions.
- Governance must be automated, observable, and owned.

Interview Questions and Model Answers

1. What is the difference between Azure Policy and Azure RBAC?

RBAC controls what actions identities can perform at scopes. Policy evaluates and enforces allowed resource state, regardless of which authorized user submits the change.

2. What are the two primary resource lock types?

CanNotDelete/Delete and ReadOnly. The first blocks deletion; the second blocks updates and deletion.

3. What is a policy initiative?

A collection of policy definitions grouped and assigned together to achieve a broader governance objective.

4. Why start policy with Audit?

It reveals existing impact and false assumptions before a Deny or Modify effect disrupts valid workloads.

Official Microsoft references: Azure Resource Manager overview -

<https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/overview> | Azure Policy overview -

<https://learn.microsoft.com/en-us/azure/governance/policy/overview> | Lock Azure resources - <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/lock-resources> | Azure Policy built-ins - <https://learn.microsoft.com/en-us/azure/governance/policy/samples/>

VOLUME 7A | CHAPTER 8

Chapter 8 - Azure Virtual Machines Foundations

Engineering secure, supportable IaaS workloads

Learning Objectives

- Explain VM images, sizes, disks, NICs, extensions, availability options, and lifecycle states.
- Select VM families based on workload characteristics rather than habit.
- Use SSH keys, managed identities, private access, diagnostics, patching, backup, and encryption.
- Identify when a managed platform is more suitable than a VM.

8.1 What an Azure VM Includes

An Azure virtual machine is an IaaS compute resource. A practical VM deployment includes an image, size, OS disk, optional data disks, network interface, subnet, network security controls, identity, boot diagnostics, monitoring, patching, backup, and availability configuration. The customer remains responsible for the guest operating system, applications, accounts, hardening, and much of recovery.

Component	Decision	Production concern
Image	Marketplace, shared image, or custom image	Patch level, provenance, hardening, and lifecycle.
Size	vCPU, memory, storage/network performance, accelerators	Quota, availability, cost, and workload fit.
OS disk	Managed disk type and encryption	Durability, performance, snapshot/backup, and key strategy.
Data disks	Separate application data from OS	LUN mapping, caching, filesystem, backup consistency.
NIC/subnet	Private IP, routing, NSG, DNS	Avoid direct broad internet exposure.
Identity	Managed identity and local admin model	No embedded cloud credentials; minimize local accounts.
Availability	Zone, availability set, or scale set	Failure isolation and update domains.
Operations	Monitor, Update Manager, Backup, Defender	Day-2 management must exist before production.

8.2 VM Size Families

Azure offers general-purpose, compute-optimized, memory-optimized, storage-optimized, GPU, and high-performance families. Exact series evolve and retire. Select a currently supported size through measured CPU, memory, disk throughput/IOPS, network bandwidth, accelerator, licensing, and cost requirements. Validate regional availability and subscription quota before committing architecture.

8.3 Availability Options

Option	Protection	Use
Single VM	No instance-level redundancy	Lab or disposable workload.
Availability set	Fault and update domain distribution within a datacenter architecture	Legacy/regional VM designs where supported.
Availability zones	Physical zone isolation inside supported region	Critical regional workloads with multi-zone architecture.
Virtual Machine Scale Sets	Fleet deployment, health, scaling, and updates	Horizontally scalable services and standardized instances.
Multi-region recovery	Regional disruption protection	Business-critical workloads with tested DR.

8.4 VM Security Baseline

- Use current supported images and automated image pipelines.
- Prefer SSH keys for Linux and tightly controlled authentication for Windows.
- Use Azure Bastion, VPN/ExpressRoute, JIT access, or restricted management sources.
- Enable managed identity for Azure access and keep secrets in Key Vault.
- Collect boot diagnostics, guest metrics/logs, security findings, and change evidence.
- Use Update Manager or an approved patch process and define maintenance windows.
- Back up data and configuration according to RPO; test restoration to a separate environment.
- Deallocate unused VMs and remove orphan disks, NICs, IPs, snapshots, and backup items when approved.

8.5 When Not to Use a VM

A VM is appropriate when the workload needs guest OS control, unsupported middleware, specialized drivers, or lift-and-shift compatibility. It may be the wrong choice when the application can use App Service, Container Apps, AKS, Functions, managed databases, or SaaS. Managed services reduce some operational work but introduce service constraints and require different architecture skills.

Hands-on Lab 8: Deploy and Harden a Linux VM

Objective Create a small Linux VM with controlled access, managed identity, monitoring evidence, and complete cleanup.

Prerequisites

- Azure subscription with VM quota in an approved region.
- Azure CLI and an SSH client.
- A source IP or approved Bastion/private-access method.
- A budget and expiry tag.

Implementation

53. Create resource group rg-v7a-vm-lab with ownership and expiry tags.
54. Create a VNet/subnet or use the networking lab design. Do not allow unrestricted inbound management access.

55. Create a small supported Linux VM with SSH keys, Standard public IP only if required, and a currently available size. Example: `az vm create --resource-group rg-v7a-vm-lab --name vm-v7a-linux --image Ubuntu2204 --size Standard_B1s --admin-username azureuser --generate-ssh-keys --assign-identity`.
56. Restrict SSH to your approved source address or use Bastion/private access.
57. Run `az vm get-instance-view` and capture power/provisioning status.
58. Inspect OS disk, NIC, NSG, managed identity, boot diagnostics, and activity log.
59. Connect and inspect OS version, disks, routes, listening ports, and cloud-init logs.
60. Deallocate the VM and compare power state with stopped state.

Validation Evidence

- VM resource ID, size, image, zone/availability decision, and identity.
- Network rule proving management is not open to the world.
- Guest evidence and activity log.
- Deallocation and cleanup evidence.

Cleanup

- Delete the resource group and verify dependent disk, NIC, public IP, and NSG removal. Remove backup items separately if created.

Stretch challenge Convert the single VM into a scale-set design behind a load balancer and define immutable image, health, patch, and rollback workflows.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
VM creation returns quota error	Regional vCPU family quota or subscription limit	Check usage and quotas and requested size family	Choose supported size/region or request quota before deployment.
SSH/RDP timeout	NSG, route, public IP, guest firewall, service, or wrong address	Use Network Watcher, effective rules/routes, boot diagnostics, and guest logs	Correct the narrowest failing layer; avoid opening broad inbound access.
Stopped VM still costs money	VM not deallocated or disks/IPs remain billable	Check power state and attached resources	Deallocate and remove unused dependent resources according to retention policy.
Extension provisioning failed	Guest agent, network egress, script error, or unsupported image	Inspect extension status and logs	Fix script/idempotency/network access and rerun safely or rebuild image.

Chapter Summary

- A VM is a system of compute, storage, network, identity, diagnostics, and operations resources.
- VM family and size selection must be measured and checked against regional quota.
- Production VMs require secure access, patching, monitoring, backup, and recovery tests.
- Managed platforms should be preferred when OS control is not a business requirement.

Interview Questions and Model Answers

1. What remains the customer's responsibility on an Azure VM?

Guest OS configuration and patching, applications, data, accounts, hardening, monitoring configuration, and workload recovery, among other responsibilities.

2. Why separate OS and data disks?

It improves lifecycle, performance tuning, backup strategy, rebuild, and data-management flexibility.

3. What is the difference between stopping and deallocating a VM?

A guest shutdown or stopped state can retain allocated compute. Deallocation releases compute allocation, though storage and other resources can still incur cost.

4. When would you use a scale set?

For standardized fleets requiring health management, scaling, rolling upgrades, and integration with load distribution.

Official Microsoft references: Azure Virtual Machines overview - <https://learn.microsoft.com/en-us/azure/virtual-machines/overview> | Azure VM sizes - <https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/overview> | VM availability options - <https://learn.microsoft.com/en-us/azure/virtual-machines/availability> | Azure Spot Virtual Machines - <https://learn.microsoft.com/en-us/azure/virtual-machines/spot-vms>

VOLUME 7A | CHAPTER 9

Chapter 9 - Azure Virtual Networking Foundations

Addressing, routing, segmentation, name resolution, and secure connectivity

Learning Objectives

- Explain VNets, subnets, NICs, private/public IPs, NSGs, route tables, DNS, peering, gateways, Private Link, and service endpoints.
- Plan nonoverlapping IP address spaces for cloud and hybrid connectivity.
- Distinguish routing controls from packet-filtering controls.
- Troubleshoot connectivity systematically from name resolution to application listener.

9.1 Virtual Network Building Blocks

Component	Purpose	Design concern
Virtual Network	Private IP boundary in an Azure region	Nonoverlapping CIDR, DNS, peering, and governance.
Subnet	Delegation and segmentation inside a VNet	Size for growth; some services require dedicated/delegated subnets.
NIC/private IP	Connects compute to subnet	IP allocation, accelerated networking, multiple NIC complexity.
Public IP	Internet-routable endpoint	Exposure, SKU, zone behavior, lifecycle, and cost.
NSG	Stateful network filtering on subnet or NIC	Rule priority, service tags, application security groups.
Route table/UDR	Custom next-hop control	Asymmetric routing, virtual appliances, propagation.
VNet peering	Private connectivity between VNets	Address overlap, transit, gateway use, and DNS.
VPN/ExpressRoute	Hybrid private connectivity	Redundancy, bandwidth, BGP, encryption, and failover.
Private Endpoint	Private IP for supported service data plane	DNS zone design and approval workflow.
Azure DNS/Private DNS	Name resolution	Split-horizon design, links, forwarding, and ownership.

9.2 Address Planning

Choose RFC 1918 or approved private address ranges that do not overlap with on-premises networks, partner networks, other VNets, or future acquisitions. Oversized flat networks waste space and weaken segmentation; undersized subnets block scale and managed-service deployment. Reserve address space for growth, hub connectivity, platform services, and disaster recovery.

Layer	Example	Rationale
Enterprise cloud range	10.64.0.0/10	Central pool governed by IP address management.
Region allocation	10.64.0.0/12	Prevents accidental cross-region overlap.
Spoke VNet	10.64.16.0/20	Enough room for workload subnets and future services.
Web subnet	10.64.16.0/24	Ingress-facing application tier.
App subnet	10.64.17.0/24	Private application tier.
Private endpoints	10.64.18.0/24	Dedicated ownership and DNS policy.

9.3 Routing and Security

Azure creates system routes, and custom user-defined routes can send traffic to a virtual appliance, virtual network gateway, internet, or other supported next hop. NSGs are stateful filters, not routing devices. A connection succeeds only when DNS, route, NSG, firewall, platform service, guest firewall, and application listener all permit the path.

```
Client -> DNS -> Route -> NSG/firewall -> Load balancer/NIC -> Guest firewall ->
Process/port
```

9.4 Private Access to Azure Services

Private endpoints place a private IP in a VNet for a supported service endpoint. They commonly require matching private DNS zones so the normal service name resolves to the private IP from intended networks. Service endpoints provide a different model by extending VNet identity to a service while the service remains on its public endpoint. Selection depends on isolation, routing, DNS, service support, and policy.

9.5 Hub-and-Spoke Foundation

Enterprises often centralize connectivity, firewall, DNS forwarding, Bastion, and shared services in hub or virtual-WAN architecture while application subscriptions use spoke VNets. This can reduce duplicated controls but creates dependency and routing complexity. Platform teams must define ownership, route propagation, egress design, DNS, observability, and failure behavior.

Hands-on Lab 9: Build and Test a Segmented VNet

Objective Create a VNet with web and application subnets, apply a restrictive NSG, and troubleshoot a deliberate connectivity failure.

Prerequisites

- Azure subscription with Network Contributor or equivalent approved permissions.
- Azure CLI and a disposable resource group.
- Two small test VMs or one VM plus Network Watcher tools, subject to budget.

Implementation

61. Create resource group rg-v7a-network-lab.
62. Create VNet vnet-v7a-lab with 10.20.0.0/16 and subnets snet-web 10.20.1.0/24 and snet-app 10.20.2.0/24.
63. Create NSGs for each subnet. Allow only the required test flow from the web subnet to the app subnet on one application port.
64. Associate NSGs with subnets and inspect effective rules.
65. Deploy test endpoints with no broad public management access, or use Network Watcher connection troubleshoot with approved resources.
66. Validate allowed traffic, then introduce one deny or wrong-port rule and capture the failure.
67. Use IP flow verify, connection troubleshoot, effective routes, NSG flow evidence where available, and guest listener checks to locate the failure.
68. Restore the correct rule and confirm service recovery.

Validation Evidence

- VNet/subnet address plan.
- NSG rule tables and effective-rule evidence.
- Failure timeline, diagnostic evidence, root cause, and recovery confirmation.

Cleanup

- Delete test VMs, public IPs, NICs, NSGs, VNet, and resource group. Verify no Bastion, gateway, or diagnostic workspace remains unless intentionally retained.

Stretch challenge Extend the design to a hub-and-spoke model with centralized firewall, private DNS resolver, Bastion, on-premises VPN, and private endpoints.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Name resolves to public address instead of private endpoint	Private DNS zone missing, not linked, or resolver path wrong	Use nslookup/dig from affected network and inspect private DNS links	Create/link correct zone and forwarding path; avoid host-file workarounds.
NSG appears correct but traffic fails	Route, guest firewall, listener, load balancer, or service policy	Inspect effective routes, Network Watcher, guest ports, and health probes	Correct the failing layer instead of adding broad allow rules.
VNet peering cannot be created	Address spaces overlap or permissions missing	Compare CIDRs and inspect role assignments on both VNets	Renumber before deployment or use approved architecture; grant least privilege.
Return traffic uses wrong path	Asymmetric UDR/virtual appliance routing	Trace both directions and inspect propagation	Correct route tables, appliance forwarding, and gateway design.

Chapter Summary

- VNets and subnets provide private address and segmentation boundaries.
- NSGs filter traffic; route tables control next hops; DNS determines destination names.
- Nonoverlapping address planning is a long-term enterprise requirement.
- Private endpoints require deliberate DNS and ownership design.
- Connectivity troubleshooting must follow the complete packet and name-resolution path.

Interview Questions and Model Answers

1. What is an NSG?

A stateful Azure network filter containing prioritized inbound and outbound rules that can be associated with subnets or network interfaces.

2. What is a user-defined route?

A custom route in a route table that changes the next hop for matching traffic.

3. Why can private endpoints fail even when the endpoint is approved?

DNS may still resolve the public address, the private DNS zone may not be linked, routes or NSGs may block access, or the service authorization may deny the client.

4. Why avoid overlapping VNet address spaces?

Overlap blocks straightforward peering and hybrid routing and creates operational complexity requiring translation or renumbering.

Official Microsoft references: Azure Virtual Network documentation - <https://learn.microsoft.com/en-us/azure/virtual-network/> | Network security groups - <https://learn.microsoft.com/en-us/azure/virtual-network/network-security-groups-overview> | Virtual network routing - <https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-udr-overview> | Azure Private Link - <https://learn.microsoft.com/en-us/azure/private-link/private-link-overview> | Azure Network Watcher - <https://learn.microsoft.com/en-us/azure/network-watcher/>

VOLUME 7A | CHAPTER 10

Chapter 10 - Azure Cost Governance and Foundation Capstone

Making ownership, budgets, optimization, and cleanup part of engineering

Learning Objectives

- Explain Azure billing and Cost Management scopes, cost analysis, budgets, alerts, exports, allocation, and Advisor recommendations.
- Distinguish cost visibility, budget notification, quota control, and hard service limits.
- Identify compute, storage, network, licensing, support, backup, and idle-resource cost drivers.
- Integrate identity, governance, VM, network, and cost controls in a foundation capstone.

10.1 Cost Is a System Property

Azure cost is produced by architecture and operations: selected SKU, running time, storage tier, transactions, data transfer, replicas, backup retention, licensing, support, logging volume, and abandoned resources. Cost optimization is not simply buying discounts. It begins with ownership, measurement, demand management, safe rightsizing, and removal of waste while preserving security and reliability.

Capability	Purpose	Important limitation
Cost analysis	Explore actual and forecasted cost by scope and dimension	Data can be delayed and must be interpreted with billing model.
Budgets	Track cost or usage thresholds and notify/action through configured integrations	A budget is not automatically a hard spending cap.
Cost alerts	Notify on budgets, anomalies, scheduled conditions, and other signals	Alert routing and ownership must be tested.
Azure Advisor	Recommendations for cost and other pillars	Recommendation requires workload context and risk review.
Pricing calculator	Estimate planned Azure use	Estimate quality depends on complete assumptions.
Reservations/savings plan	Commitment-based discounts for eligible usage	Commitment risk if demand, region, or service changes.
Tags/allocation	Attribute shared or workload cost	Tag support and governance must be consistent.
Exports/APIs	Feed FinOps reporting and analysis	Data security, retention, and schema handling are required.

10.2 Common Cost Drivers

- Compute hours, VM size, operating-system and software licenses, and whether VMs are deallocated.
- Managed disks, snapshots, backup vault retention, replicas, and unattached disks.
- Public IPs, load balancers, gateways, Bastion, firewalls, NAT, private endpoints, and data processing.
- Internet and inter-region data transfer and cross-zone patterns where applicable.
- Log ingestion, retention, archive, queries, alerts, and duplicated telemetry.

- Database provisioned capacity, backup, replicas, transaction volume, and premium features.
- Orphaned development environments, forgotten proof-of-concept resources, and overprovisioned disaster-recovery capacity.

10.3 Budget and Ownership Design

Budgets should exist at meaningful ownership scopes and include multiple thresholds, actionable recipients, and escalation. A team must know what action to take at each threshold. Automated actions must be conservative because shutting down a production service can cost more than the cloud overrun. Sandbox expiry automation is safer than automatic production deletion.

Threshold	Example response
50% of monthly budget	Confirm forecast and known projects; validate tags and allocation.
75%	Review top drivers and idle resources with workload owner.
90%	Escalate forecast, approve optimization plan, and control nonessential growth.
100%+	Incident-style review for anomaly, unplanned demand, or budget error; preserve service safety.

10.4 Optimization Order

69. Eliminate resources with no owner, purpose, or required retention.
70. Schedule or deallocate nonproduction compute safely.
71. Rightsize using measured performance and business headroom.
72. Optimize storage tier, snapshots, backup retention, and telemetry volume.
73. Review architecture for data-transfer and managed-service efficiency.
74. Only then evaluate commitments such as reservations or savings plans for stable eligible demand.
75. Validate post-change reliability, security, and performance and record realized savings.

10.5 Foundation Capstone Architecture

```

Microsoft Entra groups + MFA/PIM
|
Management group / subscription guardrails
|
Resource group: governed tags + budget + lock strategy
|
VNet 10.30.0.0/16
+-- management/private access
+-- web subnet -> Linux VM with managed identity
+-- application subnet reserved for later parts
|
Activity logs + cost analysis + cleanup evidence

```

Hands-on Lab 10: Azure Foundation Deployment and Operations Review

Objective Design, deploy, secure, observe, cost-control, test, and remove a small Azure foundation environment.

Prerequisites

- Completed Chapters 1-9 or equivalent knowledge.
- Azure sandbox subscription with approved VM and network quota.
- Budget, naming standard, and expiry date.
- Permission to create resources and limited role assignments.

Implementation

76. Write requirements: owner, environment, users, workload identity, region, RTO/RPO, data class, budget, allowed management path, and cleanup date.
77. Validate tenant and subscription context and create resource group rg-v7a-capstone with required tags.
78. Create a VNet with management, web, and application subnets using a nonoverlapping address plan.
79. Deploy one small Linux VM in the web subnet with SSH key authentication, managed identity, diagnostics, and no unrestricted inbound management access.
80. Grant the VM identity a minimal Reader role on the resource group and explain why this is still broader than many production workloads need.
81. Apply an approved lock strategy and an audit policy for one governance requirement.
82. Create a Cost Management budget at the permitted scope with testable notification thresholds, or document the exact portal configuration if account type restricts creation.
83. Generate activity by starting, stopping, deallocating, and inspecting the VM. Capture activity log, role assignment, effective NSG rules, resource IDs, and cost views.
84. Inject one failure: remove required network rule, use wrong subscription, or attempt locked deletion. Diagnose and recover through evidence.
85. Present an operations review covering security, reliability, observability, cost, known gaps, and next steps.
86. Remove role assignments, locks, policy assignments, budget where appropriate, VM resources, and the resource group. Verify no orphan resource remains.

Validation Evidence

- Architecture diagram and decision record.
- CLI transcript with context checks and no secrets.
- Identity and RBAC evidence.
- Network and failure-troubleshooting evidence.
- Budget/cost evidence and cleanup checklist.
- Post-lab review with lessons and improvements.

Cleanup

- Confirm deletion through Resource Graph, resource-group list, disks, NICs, public IPs, snapshots, role assignments, locks, policy assignments, and budget scope. Preserve only sanitized portfolio evidence.

Stretch challenge Rebuild the entire capstone with Bicep in a later volume, include what-if review, deployment validation, policy tests, and a CI pipeline using workload identity federation.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Budget alert never arrives	Threshold not reached, wrong recipients, permissions, action group, or processing delay	Review budget status, alert history, recipient configuration, and billing scope	Correct routing and run a controlled test; maintain alternate FinOps monitoring.
Cost appears after resource deletion	Usage reporting delay or retained disks/backups/network resources	Inspect cost by resource and inventory dependent resources	Wait for reporting latency and remove only resources approved for deletion.
Advisor suggests a risky shutdown/rightsize	Recommendation lacks workload context	Review utilization period, business calendar, resilience, and dependency data	Validate with owner and controlled test; document accepted or rejected recommendation.
Capstone cleanup blocked	Locks, backup retention, policy, permissions, or dependency	List locks, assignments, backup items, and deployment operations	Remove controls in approved order and verify each dependent resource.

Chapter Summary

- Cost governance requires architecture, ownership, measurement, budgets, alerts, and cleanup.
- Budgets notify and can integrate with actions, but they are not inherently hard spending caps.
- Optimization must preserve security, reliability, and business performance.
- The Part A capstone integrates tenant context, hierarchy, identity, governance, compute, networking, troubleshooting, and cost.

Interview Questions and Model Answers

1. What is the difference between a budget and a quota?

A budget tracks cost or usage and sends alerts or triggers configured actions. A quota limits a service quantity or capacity; it is not a financial budget.

2. Why can a deleted VM still generate cost?

Managed disks, snapshots, backup retention, public IPs, networking services, or delayed usage reporting can remain.

3. When should reservations or savings plans be considered?

After demand is measured, waste is removed, architecture is stable, and commitment risk is understood.

4. How do tags support FinOps?

They help attribute cost to owners, environments, applications, and cost centers, enabling showback, accountability, automation, and anomaly investigation.

Official Microsoft references: Cost Management overview - <https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/overview-cost-management> | Create and manage budgets - <https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-create-budgets> | Plan to manage Azure costs - <https://learn.microsoft.com/en-us/azure/cost-management-billing/understand/plan-manage-costs> | Cost optimization best practices - <https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/cost-mgt-best-practices> | Azure Advisor cost recommendations - <https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-opt-recommendations>

Part A Command Reference

Task	Azure CLI command
Show active context	<code>az account show --output jsonc</code>
List subscriptions	<code>az account list --output table</code>
Set subscription	<code>az account set --subscription "<id-or-name>"</code>
Create resource group	<code>az group create --name <rg> --location <region> --tags owner=<owner> environment=lab</code>
List resource groups	<code>az group list --query "[].{name:name,location:location}" --output table</code>
Show resource group	<code>az group show --name <rg> --output jsonc</code>
Create VNet and subnet	<code>az network vnet create --resource-group <rg> --name <vnet> --address-prefix 10.20.0.0/16 --subnet-name <subnet> --subnet-prefix 10.20.1.0/24</code>
Create managed identity	<code>az identity create --resource-group <rg> --name <identity></code>
List role assignments	<code>az role assignment list --assignee <object-id> --all --output table</code>
Create delete lock	<code>az lock create --name <lock> --lock-type CanNotDelete --resource-group <rg></code>
List locks	<code>az lock list --resource-group <rg> --output table</code>
Create Linux VM	<code>az vm create --resource-group <rg> --name <vm> --image Ubuntu2204 --admin-username azureuser --generate-ssh-keys --assign-identity</code>
View VM status	<code>az vm get-instance-view --resource-group <rg> --name <vm> --output table</code>
Deallocate VM	<code>az vm deallocate --resource-group <rg> --name <vm></code>
Recent activity log	<code>az monitor activity-log list --resource-group <rg> --max-events 20 --output table</code>
Delete resource group	<code>az group delete --name <rg> --yes --no-wait</code>

Command warning Commands and aliases can change. Check current Azure CLI help, API behavior, regional availability, and organization policy before execution. Human examples are not production automation without identity, review, idempotency, testing, and error handling.

Production Readiness Checklist - Azure Foundations

Domain	Readiness evidence
Identity	MFA and Conditional Access; privileged roles eligible/time-bound; emergency access tested; workload identities avoid stored secrets.

Domain	Readiness evidence
Hierarchy	Management-group and subscription model documented; ownership and delegation clear; resource groups follow lifecycle.
Authorization	Group-based Azure RBAC; least privilege; management/data-plane roles distinguished; access reviews scheduled.
Governance	Naming, tags, policy initiatives, exemptions, locks, provider registration, and deployment standards controlled as code.
Region/reliability	Region and zone support verified; quotas approved; RTO/RPO and failure tests documented.
Compute	Supported image/size; patching, backup, diagnostics, identity, secure access, and deallocation lifecycle defined.
Networking	Nonoverlapping CIDR; routing and NSG ownership; DNS/private endpoint plan; hybrid growth reserved.
Observability	Activity logs, diagnostic settings, operational alerts, and audit retention defined.
Cost	Budget, alert routing, tags/allocation, owner, expiry, Advisor review, and orphan-resource process operating.
Automation	Context validation, workload identity, structured output, idempotency, version testing, and code review used.
Cleanup/recovery	Deletion order, locks, backup retention, restoration tests, and sanitized evidence process documented.

Cross-Chapter Troubleshooting Method

87. Confirm the Microsoft Entra tenant, signed-in identity, and active subscription.
88. Capture the full error, timestamp, correlation/request ID, resource ID, operation, and client tool version.
89. Check Azure RBAC, PIM activation, deny assignments, Conditional Access, and token freshness.
90. Check management-group/subscription policy assignments, locks, provider registration, quotas, region, SKU, and API version.
91. For network issues, verify DNS, route, NSG/firewall, platform endpoint, guest firewall, and process listener in order.
92. For VM issues, inspect provisioning state, instance view, boot diagnostics, extensions, guest agent, disk state, and activity log.
93. For cost issues, identify scope, billing period, resource ID, meter, tags, retained dependencies, and reporting latency.
94. Make the smallest reversible correction, validate the business transaction, and record evidence and prevention action.

Glossary

Term	Definition
Azure Resource Manager (ARM)	Azure management and deployment layer that organizes resources and routes management operations to resource providers.

Term	Definition
Availability zone	Separated datacenter group within a supported region with independent power, cooling, and networking.
Management group	Scope used to organize subscriptions and apply governance and access at scale.
Subscription	Azure resource, quota, billing, and governance boundary associated with a Microsoft Entra tenant.
Resource group	Lifecycle and management container for Azure resources.
Microsoft Entra ID	Cloud identity and access service used for users, groups, applications, devices, and authentication controls.
Azure RBAC	Authorization system assigning role definitions to principals at Azure resource scopes.
Managed identity	Azure-managed workload identity used to obtain Microsoft Entra tokens without storing credentials in code.
PIM	Privileged Identity Management, providing eligible, time-bound, controlled privileged access.
Azure Policy	Service for evaluating and enforcing organizational standards on Azure resource state.
Resource lock	Delete or Read-only protection applied to subscription, resource group, or resource scope.
VNet	Azure Virtual Network, a private network boundary in an Azure region.
NSG	Network security group, a stateful rule set filtering inbound and outbound network traffic.
UDR	User-defined route that changes next-hop behavior for matching traffic.
Private endpoint	Private IP in a VNet providing access to a supported service through Azure Private Link.
RTO	Recovery time objective: maximum acceptable restoration time.
RPO	Recovery point objective: maximum acceptable data loss measured in time.
FinOps	Collaborative practice connecting engineering, finance, and business ownership to cloud value and cost decisions.

Official Microsoft Documentation Index

Azure documentation: <https://learn.microsoft.com/en-us/azure/>

Azure reliability: <https://learn.microsoft.com/en-us/azure/reliability/>

Azure Resource Manager: <https://learn.microsoft.com/en-us/azure/azure-resource-manager/>

Microsoft Entra: <https://learn.microsoft.com/en-us/entra/>

Azure RBAC: <https://learn.microsoft.com/en-us/azure/role-based-access-control/>

Azure Policy: <https://learn.microsoft.com/en-us/azure/governance/policy/>

Azure Cloud Shell: <https://learn.microsoft.com/en-us/azure/cloud-shell/>

Azure CLI: <https://learn.microsoft.com/en-us/cli/azure/>

Azure Virtual Machines: <https://learn.microsoft.com/en-us/azure/virtual-machines/>

Azure Virtual Network: <https://learn.microsoft.com/en-us/azure/virtual-network/>

Cost Management and Billing: <https://learn.microsoft.com/en-us/azure/cost-management-billing/>

Cloud Adoption Framework: <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/>

Azure Architecture Center: <https://learn.microsoft.com/en-us/azure/architecture/>

Documentation currency Azure services, names, limits, prices, regional availability, defaults, licensing, and retirement dates change. Treat official Microsoft documentation, Azure Service Health, product lifecycle notices, and organization standards as the production source of truth.

Part A Completion Record

- Complete all ten labs or approved equivalents.
- Pass a practical review covering tenant/subscription context, Azure RBAC, managed identity, policy, locks, VM operations, VNet troubleshooting, and cost cleanup.
- Present the foundation capstone and demonstrate one controlled failure and recovery.
- Submit sanitized evidence, architecture decisions, and lessons learned.
- Proceed to Volume 7B only after the foundation environment can be created and removed safely.

Next part Volume 7B will deepen Azure compute, networking, load balancing, DNS, hybrid connectivity, storage, backup, and disaster-recovery engineering.